

Question number	Answer	Additional guidance	Mark
5(a)	maintaining the body in a state of dynamic equilibrium (1)	ACCEPT maintenance of a constant internal environment ACCEPT {control / regulation} of a constant internal environment IGNORE example of homeostasis described	(1)

Question number	Answer	Additional guidance	Mark
5(b)	An answer that includes the following points: Similarities - they both decrease (over the duration of exercise) (1) - they both show the faster decrease over the first 50 minutes / slower decrease after 50 minutes (1) Differences - the concentration of insulin decreases {more / faster} in the untrained individual / after 50 mins insulin concentration decreases in trained individuals slower than untrained individuals / (apart from 0) trained individuals always have {higher / less negative} insulin concentration than untrained individuals (1)	Max 2 similarities IGNORE reference to time for mp1 IGNORE levels off after 50 mins ACCEPT gradient {described / calculated} for comparison trained / untrained ACCEPT converse	(3)

Question number	Answer	Additional guidance	Mark
5(c)(i)	A description that includes three of the following points: • {extract / isolate / cut} the {two genes / genes for both polypeptides} using {restriction enzyme / endonuclease} (1) • cut plasmid with same {restriction enzyme / endonuclease} (1) • {splice / join/insert} {sticky ends / plasmid and gene(s)} together with (DNA) ligase (1) • plasmid inserted into bacteria (1)	IGNORE restricting enzyme IGNORE restricting enzyme IGNORE gene being inserted into bacterium	(3)

Question number	Answer	Additional guidance	Mark
5(c)(ii)	An answer that includes two of the following points: • not limited by the slaughter of animals (1) • {large / more} quantities can be made / cheaper (1) • no {ethical / religious} (issues linked to the {use of / killing} animals) (1) • more effective at treating diabetes / reduced {risk of rejection / allergic reaction/ immune response} (1) • no risk of transferring infections (between other animals to humans) (1)	ACCEPT animals not lost from food chain ACCEPT produced faster / produce many IGNORE cheap unqualified / bacteria produced faster ACCEPT animal insulin may be different so less effective	(2)

Question number	Answer	Additional guidance	Mark
5(d)(i)	An answer that includes one of the following points: - no {change / increase} (in the number of cells) (1) - cells will not {grow / divide (by mitosis) without insulin (1)		(1)

Question number	Answer	Additional guidance	Mark
5(d)(ii)	A description that includes two of the following points: - contained repeat (each concentration / experiment) / {mean(s) / average(s)} calculated (1) (so that) standard {deviation(s) / error(s)} can be calculated (1)	ACCEPT replicates ACCEPT SDs calculated	(2)

Question number	Answer	Additional guidance	Mark
5(d)(iii)	Choose an item • {size / overlap} of error bars compared (1) • Overlapping error bars shows no significant difference / non overlapping error bars show significant difference (1)	IGNORE reliability / validity / repeatability	(1)